CLUB BCI / IMPLEMENTATION ATLAS

Visual maps

System structure, data, functions, user journeys and build pipelines

26 rendered diagrams, traced from the working tree on November 11, 2026. Each diagram has its own vector page. Use the PDF bookmarks or page numbers below to jump to a topic. Zoom in for detail.

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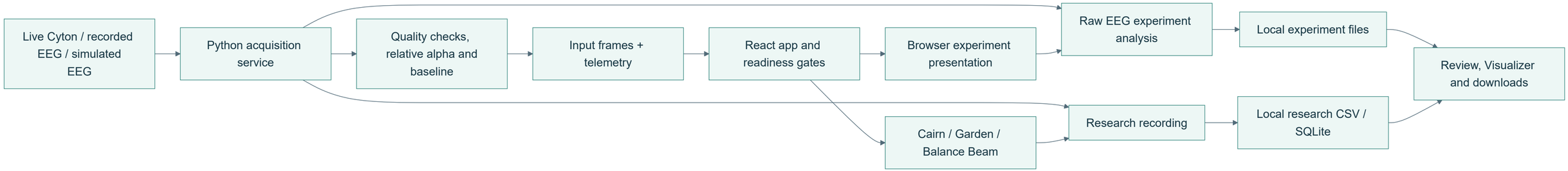
3 diagrams: Install, run and build / Version guard and CI paths / Build information shown in the header

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3 diagrams: Recording conversion and no-headset game testing / Cairn photo assets / Research export inspection

Source: docs/workspace/maps / Mermaid originals remain editable

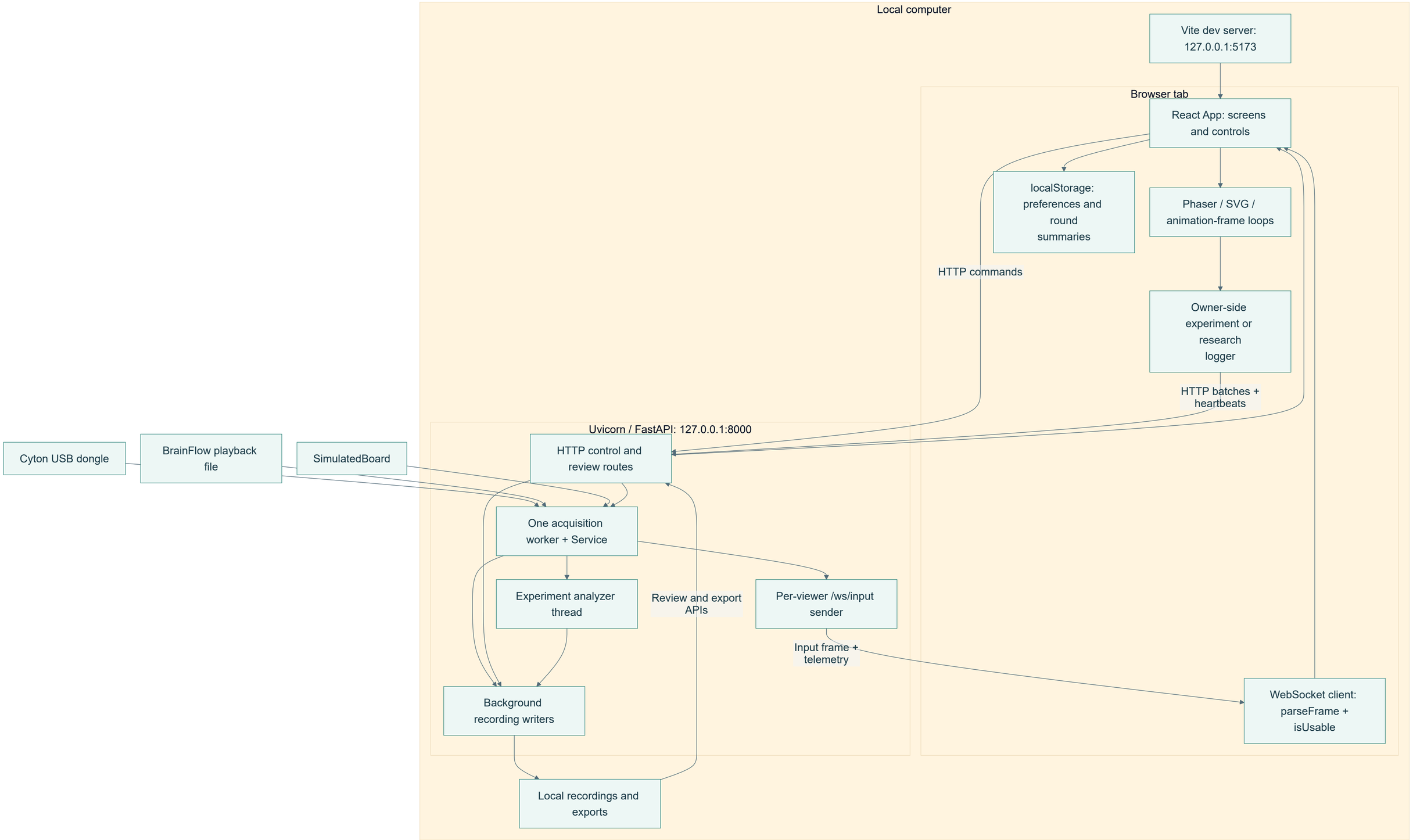
System overview



There is also a score-only synthetic source: it bypasses raw EEG acquisition and baseline collection, so it can drive the games but cannot start an EEG experiment. No source is substituted automatically.

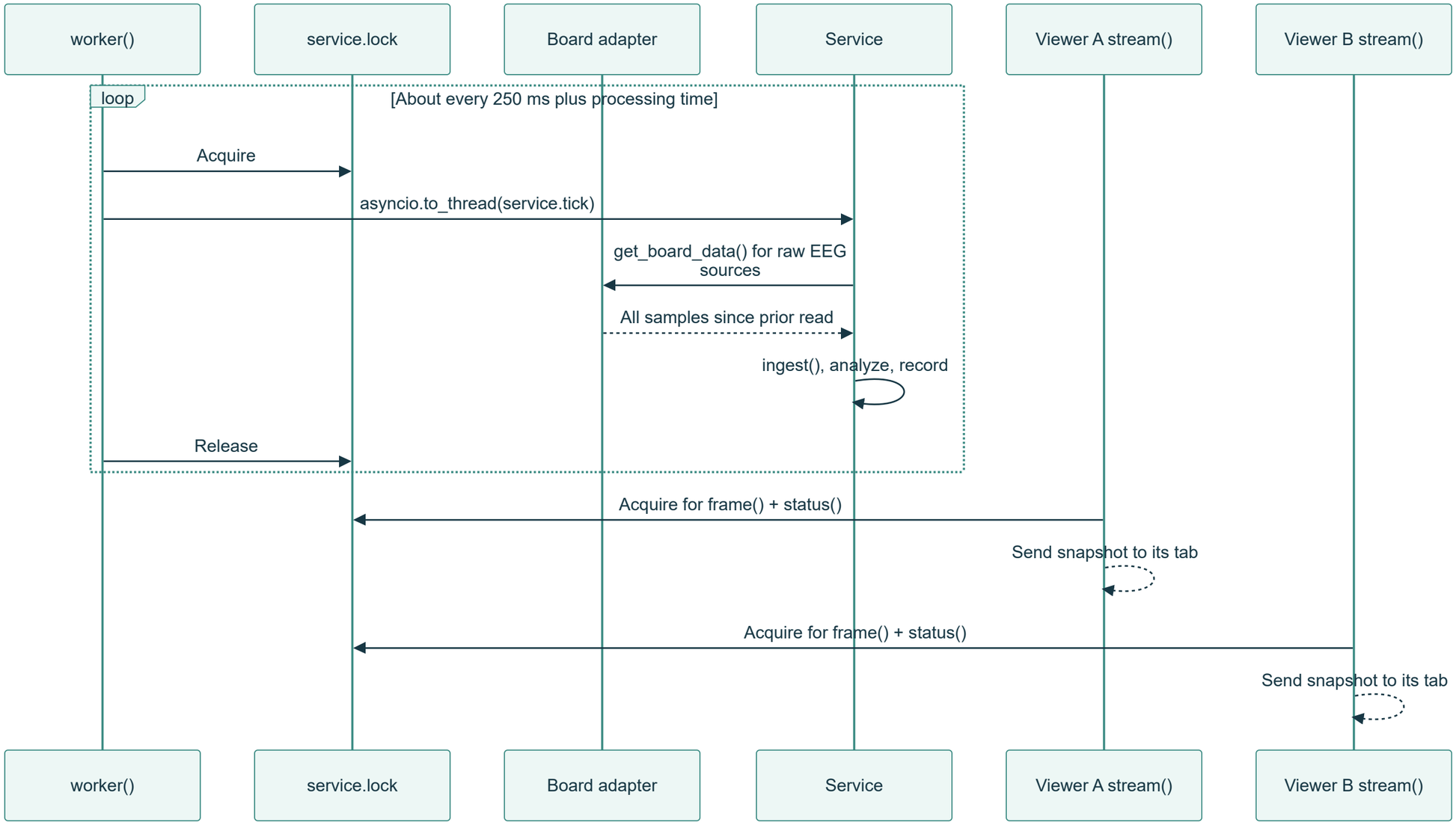
Source: docs/workspace/maps/README.md

Processes and connections



Vite serves the frontend in development. The frontend build produces static assets; Python remains a separately started backend. Browser requests name port 8000 directly; this app does not use a Vite API proxy.

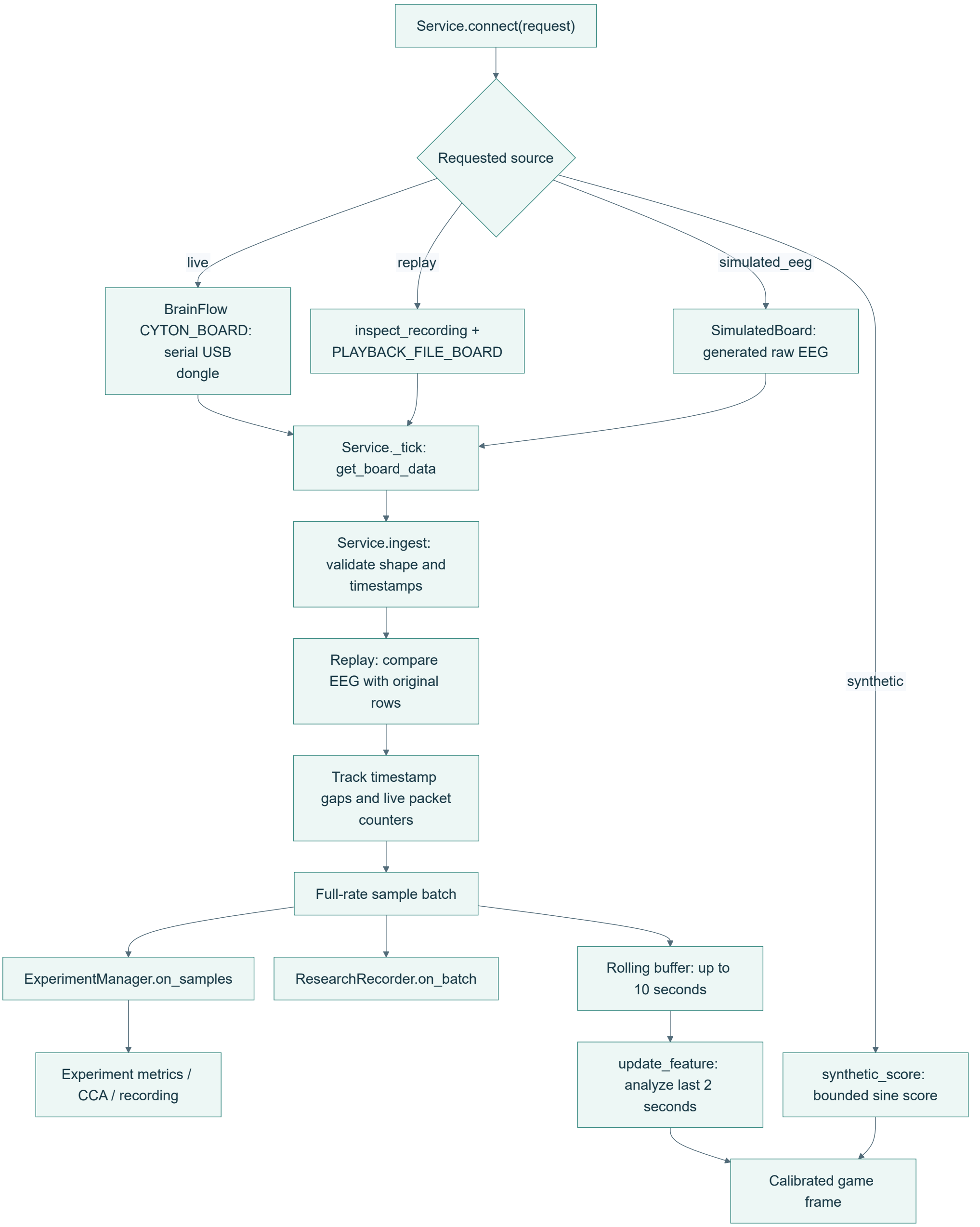
One producer, multiple viewers



> Each viewer polls snapshots at its own nominal 4 Hz cadence. A viewer never reads the board. Sequence numbers belong to the producer; the browser ignores repeated or out-of-order frames. An open WebSocket does not mean the headset is connected or the input is usable.

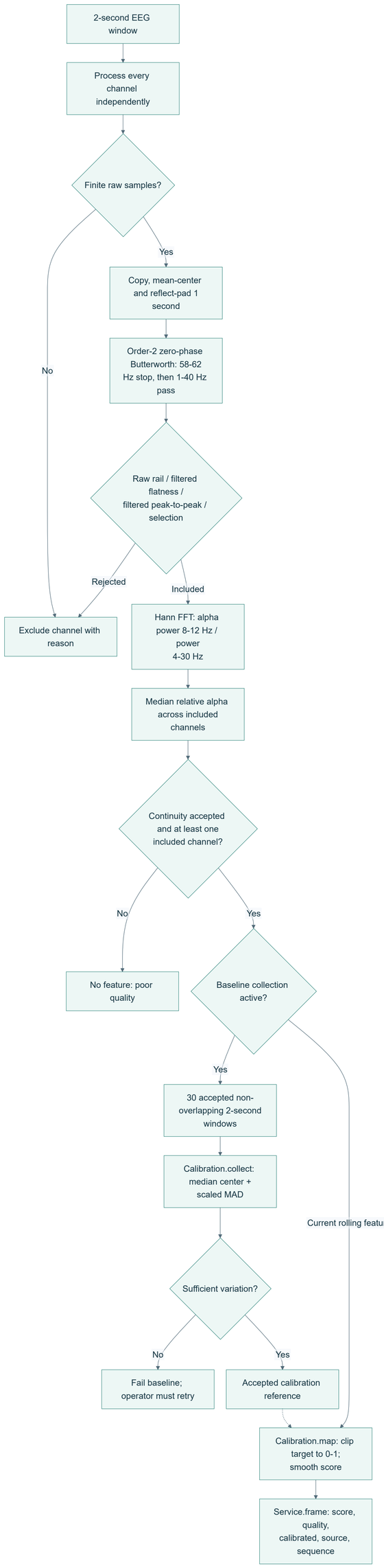
Source: docs/workspace/maps/system/runtime.md

Acquisition and fan-out



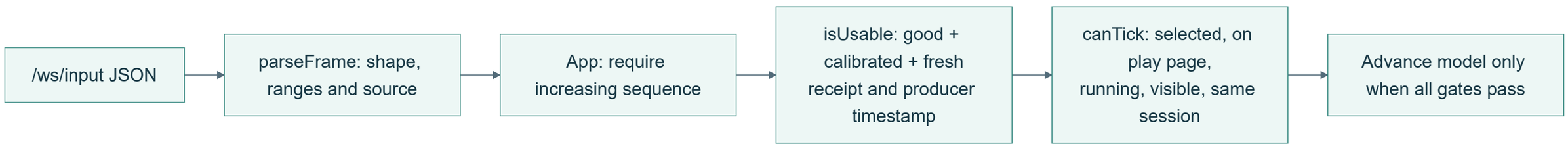
Cyton's configured layout is eight EEG channels at 250 Hz. SDK reads arrive in batches; their sizes are not assumed to equal a feature window. Full-rate recording and experiment analysis branch before the game amplitude gate. Chart telemetry is a reduced view of the rolling buffer, not the samples used for decoding or recording.

Per-window processing and calibration



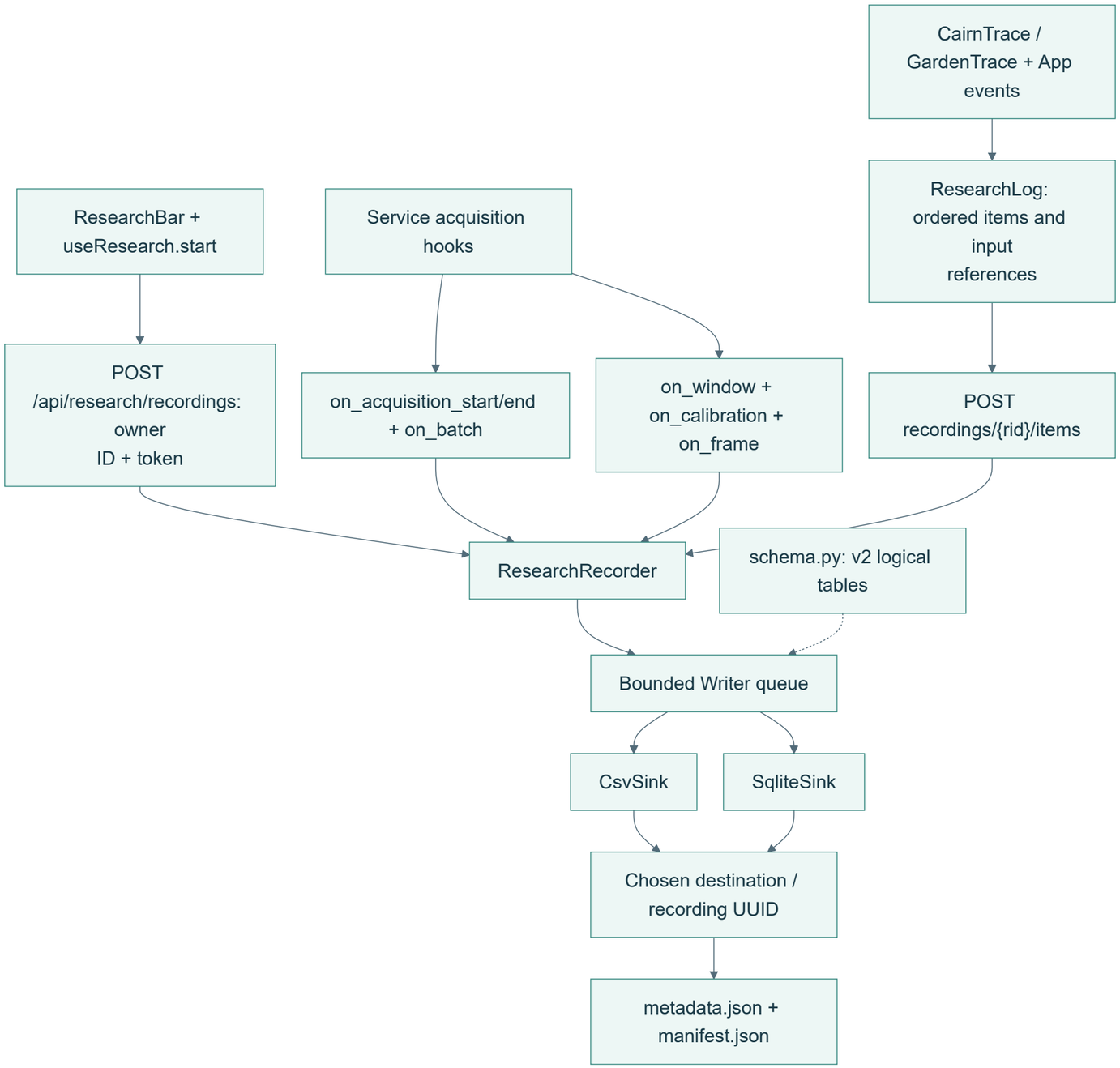
Baseline collection and the rolling play feature share analyze(); the diagram separates their consumers. Baseline time means accepted data: rejected windows do not advance the 60-second target. Live EEG requires an explicit scoring-channel selection before calibration. Raw mains-hum RMS is measured separately for diagnostics; the notch filter does not establish electrode contact.

Browser acceptance



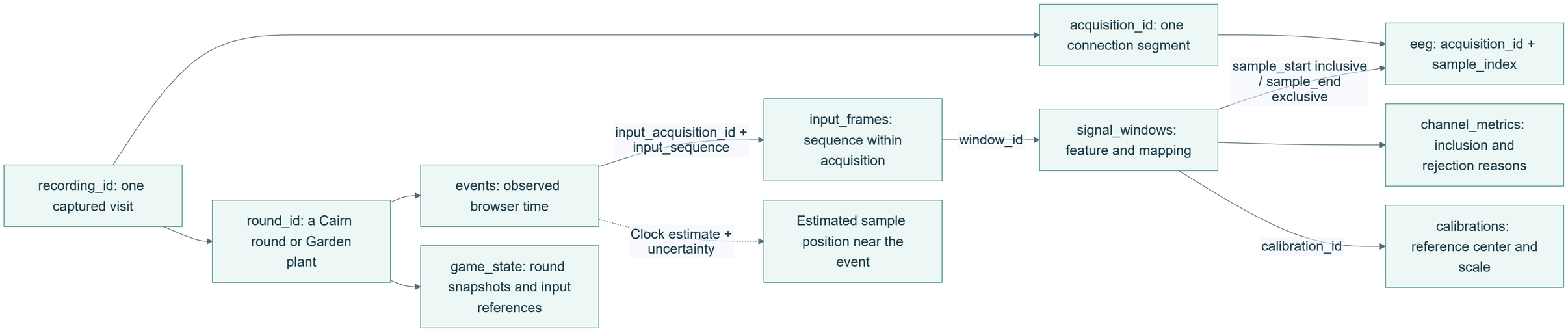
> isUsable() checks both the monotonic receipt age and the producer's wall-clock timestamp against a two-second limit; it also rejects a producer timestamp over two seconds in the future. Acquisition freshness is checked separately by the backend. Recovery of usable input does not press Resume for the player.

Research / Dev Mode capture



> The selected format creates CSV, SQLite, or both. The default destination is data/research/; BCI_RESEARCH_DIR and configured destinations can change it. Recording may begin before connection or partway through a session and can span reconnects, recalibration and game switches. Beginning mid-session is marked as partial rather than inventing earlier data.

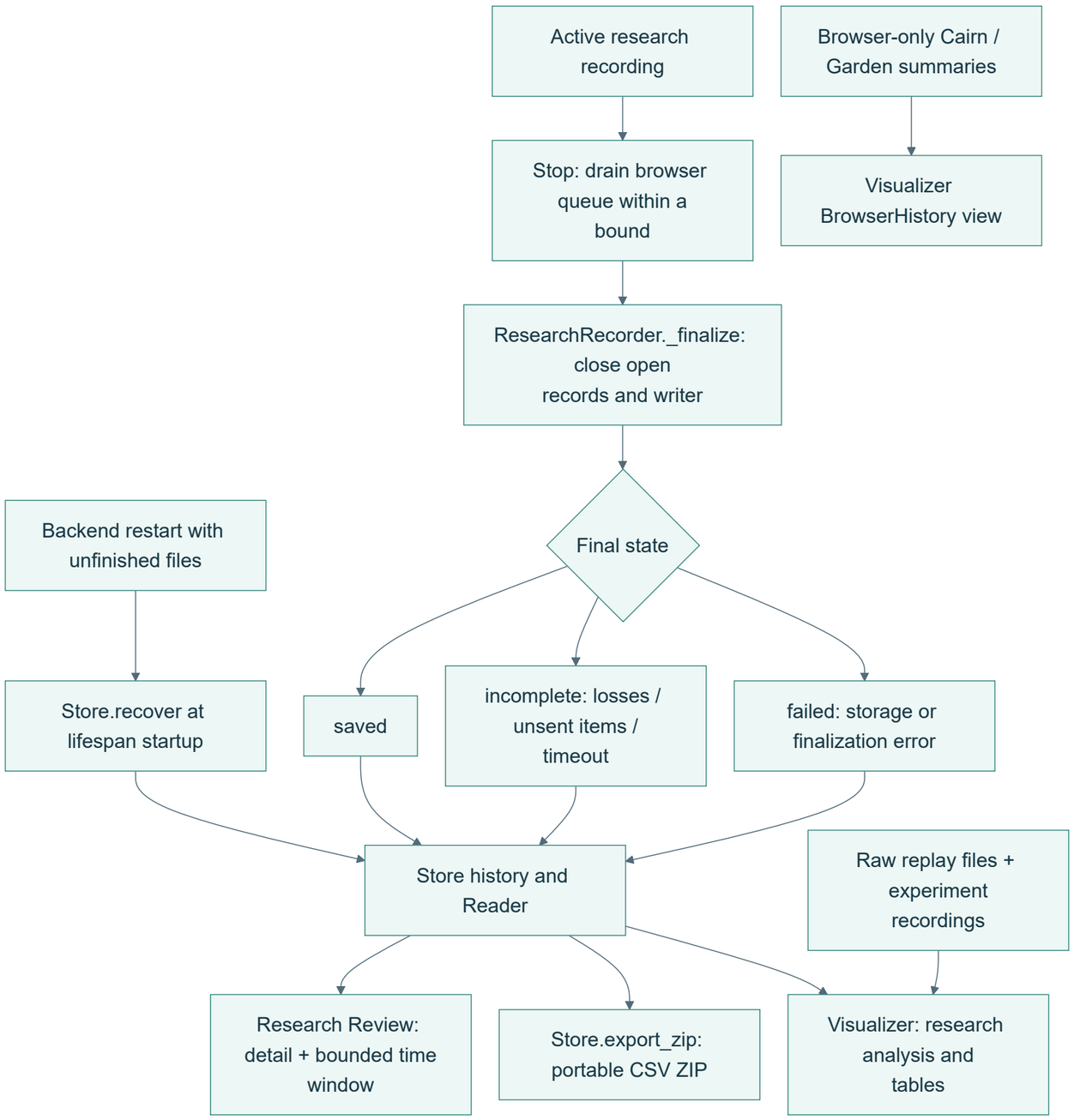
Trace one event back to EEG



> These are logical joins in the recorded schema, not a claim that every storage format enforces foreign keys. The input-frame reference identifies the score the game consumed. Clock-aligned sample estimates identify nearby raw data with uncertainty; browser event times are not exact hardware sample triggers. Score-only synthetic frames have no raw EEG window.

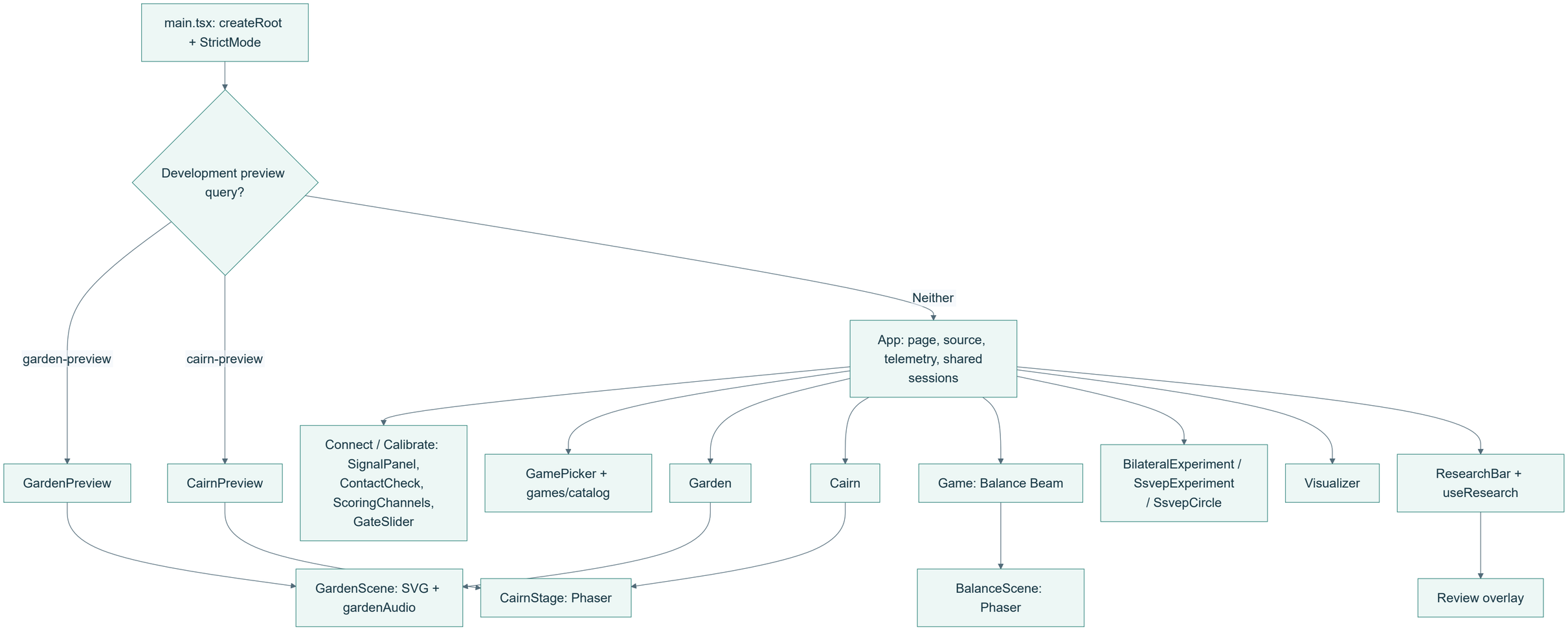
Source: docs/workspace/maps/data/recording-and-review.md

Save, recover and inspect



> Stop does not stop acquisition or automatically end ordinary gameplay. A recording finalizes asynchronously; file completion and losses determine its final state. Owner loss after five seconds or a full storage queue ends capture with an explicit incomplete/failed outcome. Export reads persisted files; SQLite-only research recordings can still export CSV ZIPs. Inspect active capture in Live rather than treating it as a finalized recording.

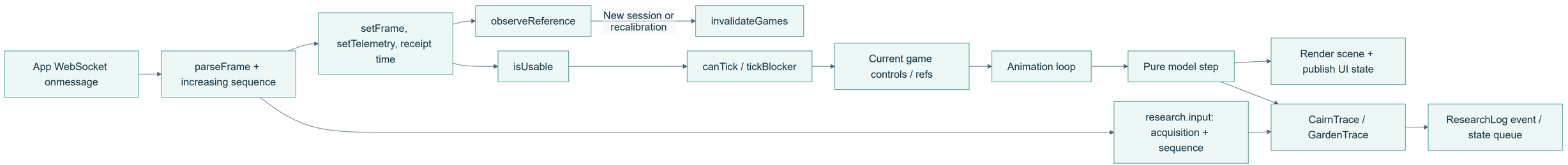
Entry points and component ownership



The normal page state is setup, calibrate, choose, play or visualizer. The initial #visualizer hash opens Visualizer directly. Research controls and Review are overlays around this flow, rather than separate acquisition sessions. Dev preview queries are enabled only when import.meta.env.DEV is true.

Source: docs/workspace/maps/code/frontend.md

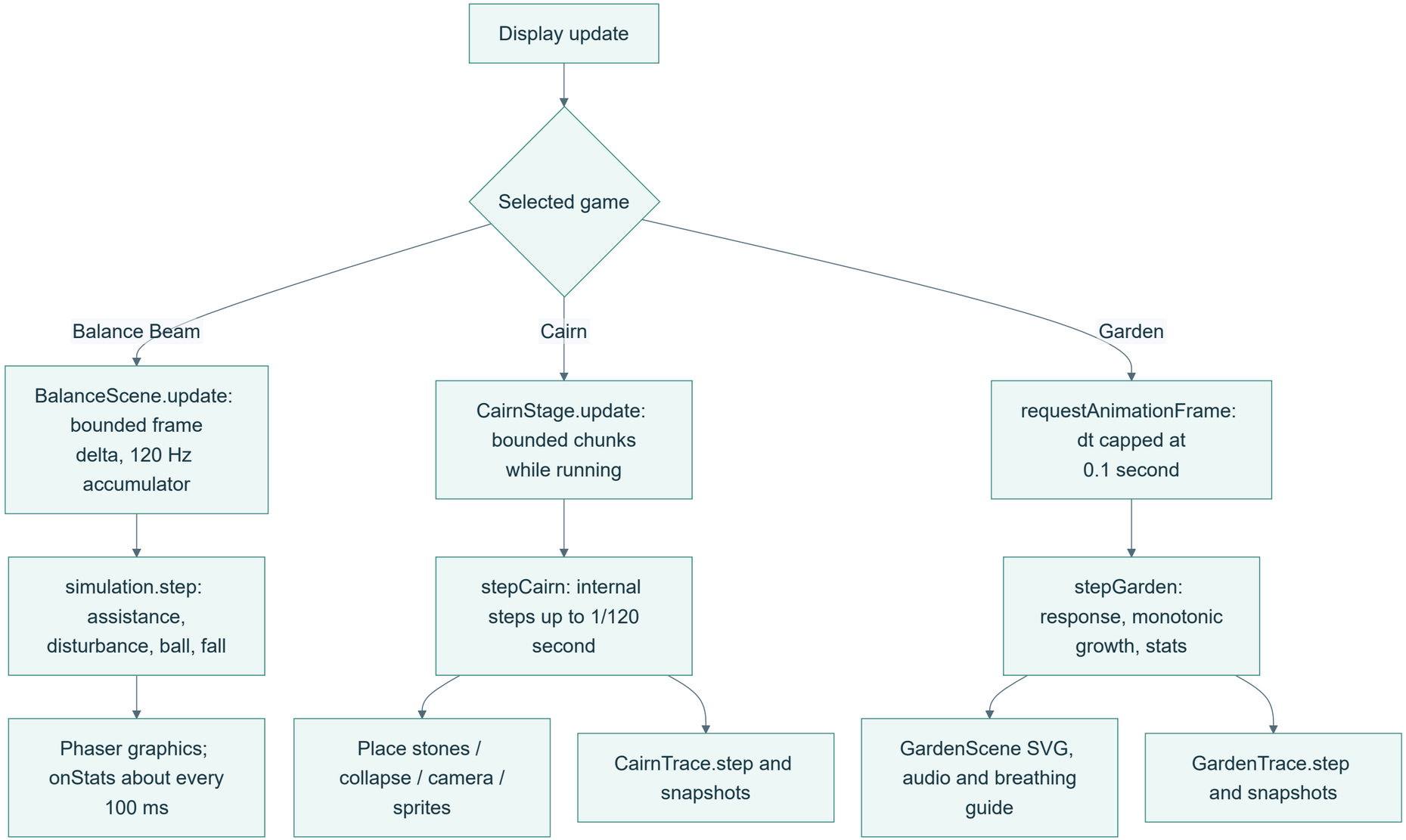
From a network frame to a rendered game



React owns navigation and controls. Mutable refs hold the latest controls/model for animation callbacks. A missing telemetry update is treated as an outage; an observed new session or transition into calibration invalidates rounds tied to the old comparison reference.

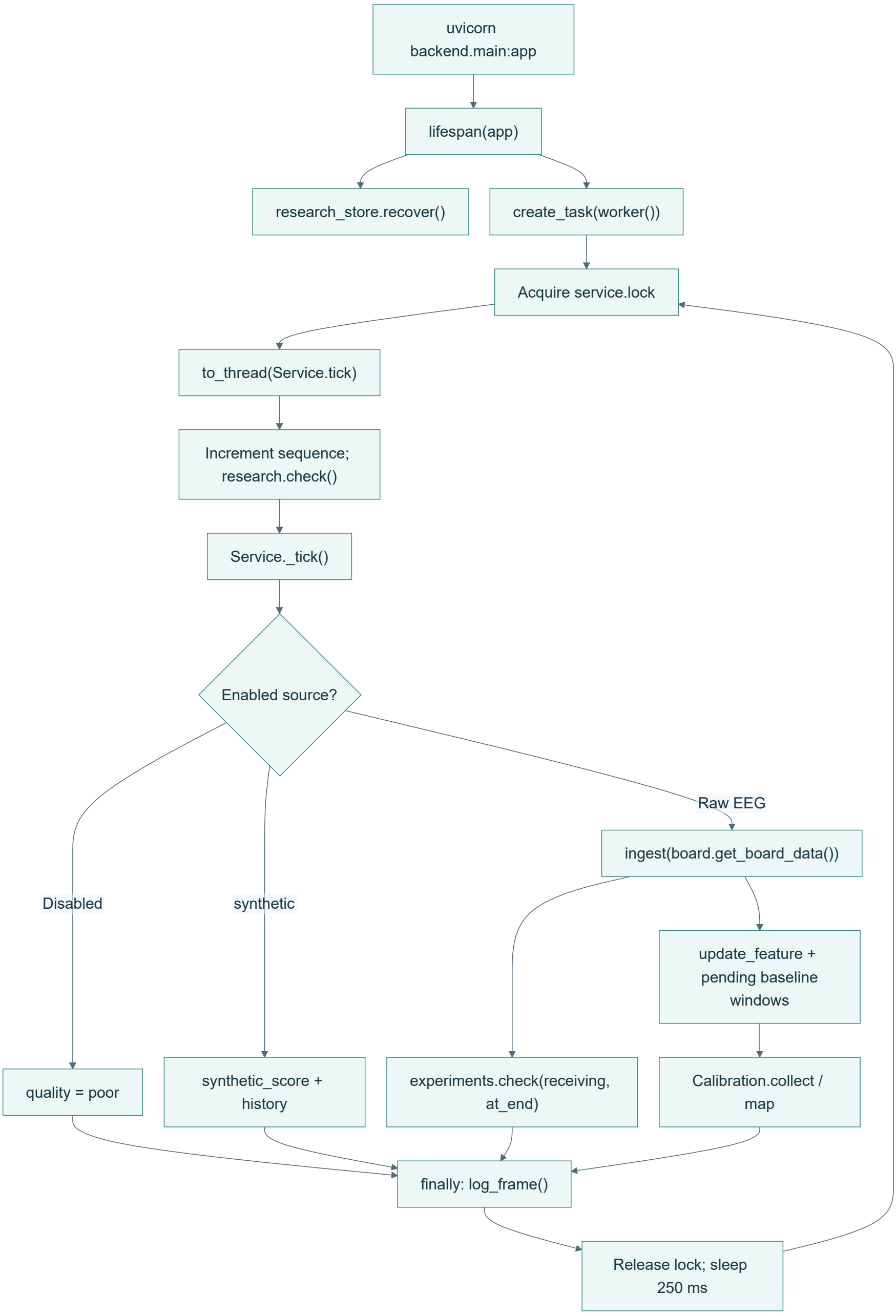
Source: docs/workspace/maps/code/frontend.md

Game update loops



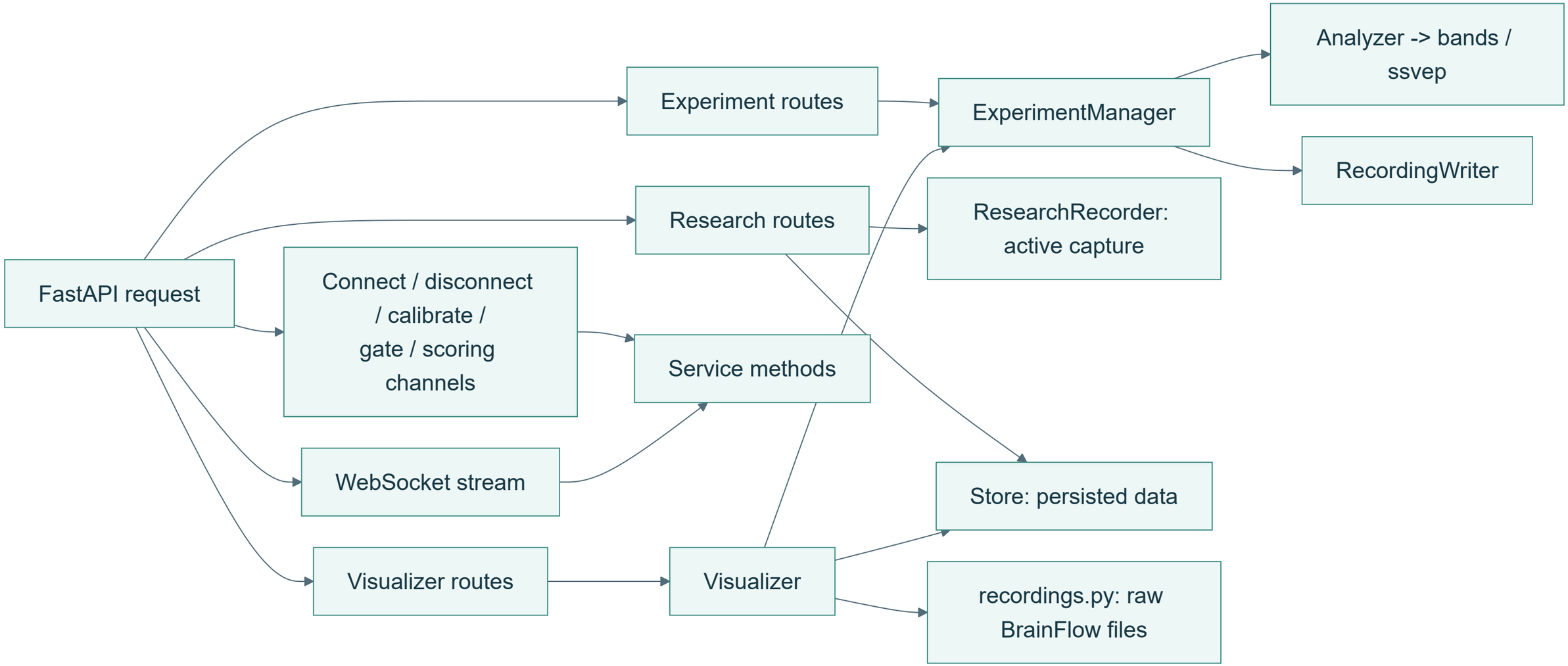
Paused game progress does not catch up after a background-tab delay. Presentation can still finish a fall animation or settle an ambient visual; those animations do not imply gameplay time is advancing. Research state snapshots for Cairn/Garden target 20 Hz rather than copying every display frame. Balance Beam consumes the shared score but has no equivalent dedicated game-state trace class in this implementation.

Startup and acquisition tick



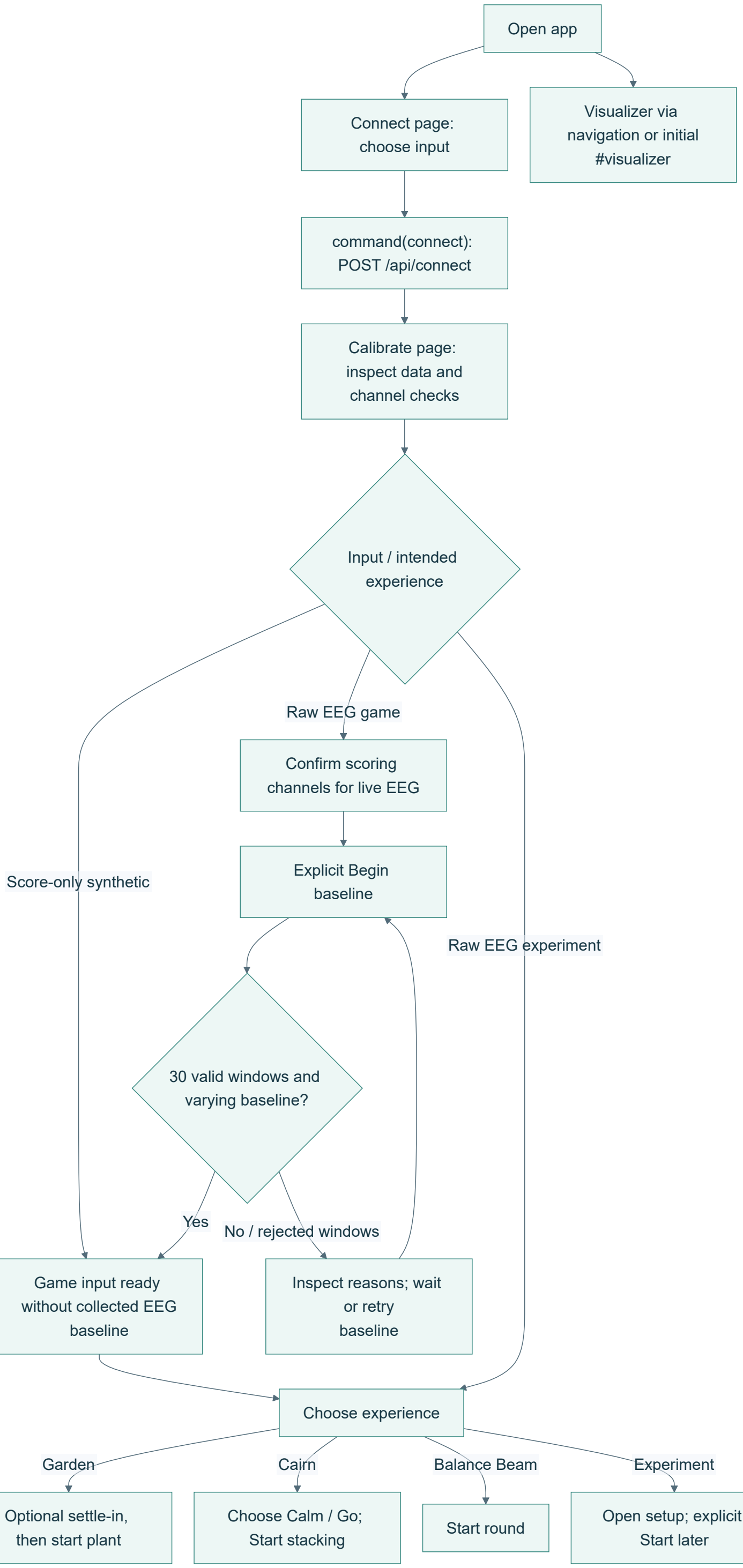
The worker catches acquisition exceptions, records the error and calls Service.stop(). Tick logging is in a finally block so unusable frames remain visible in research capture. The HTTP control routes also use the service lock when changing acquisition or baseline settings. stream() sends snapshots independently of the worker, as shown in the runtime map.

Request dispatch



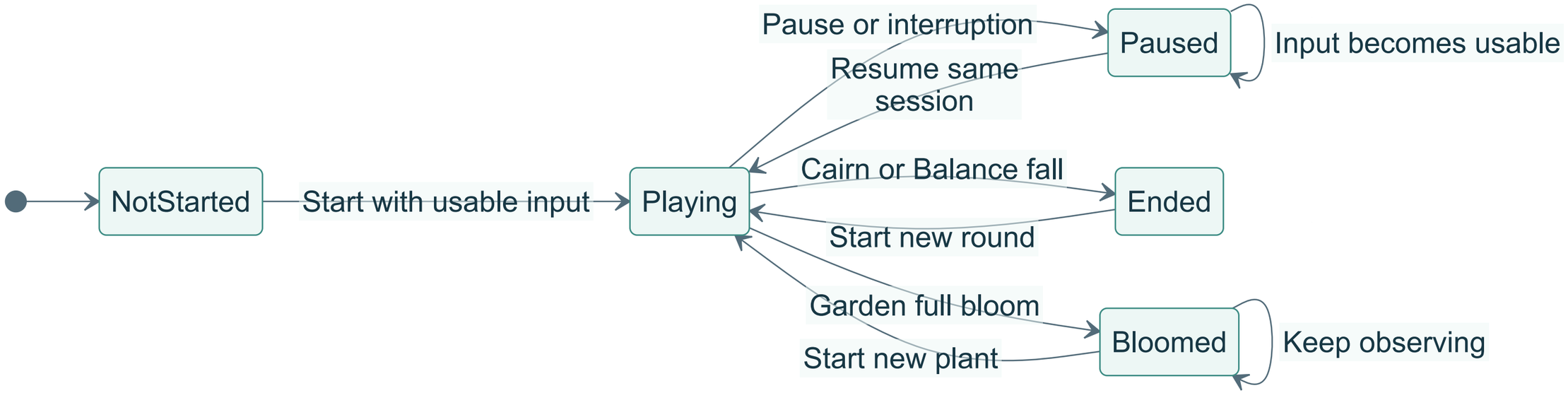
Follow the labeled arrows and source references in the companion Markdown guide.

Connect to an experience



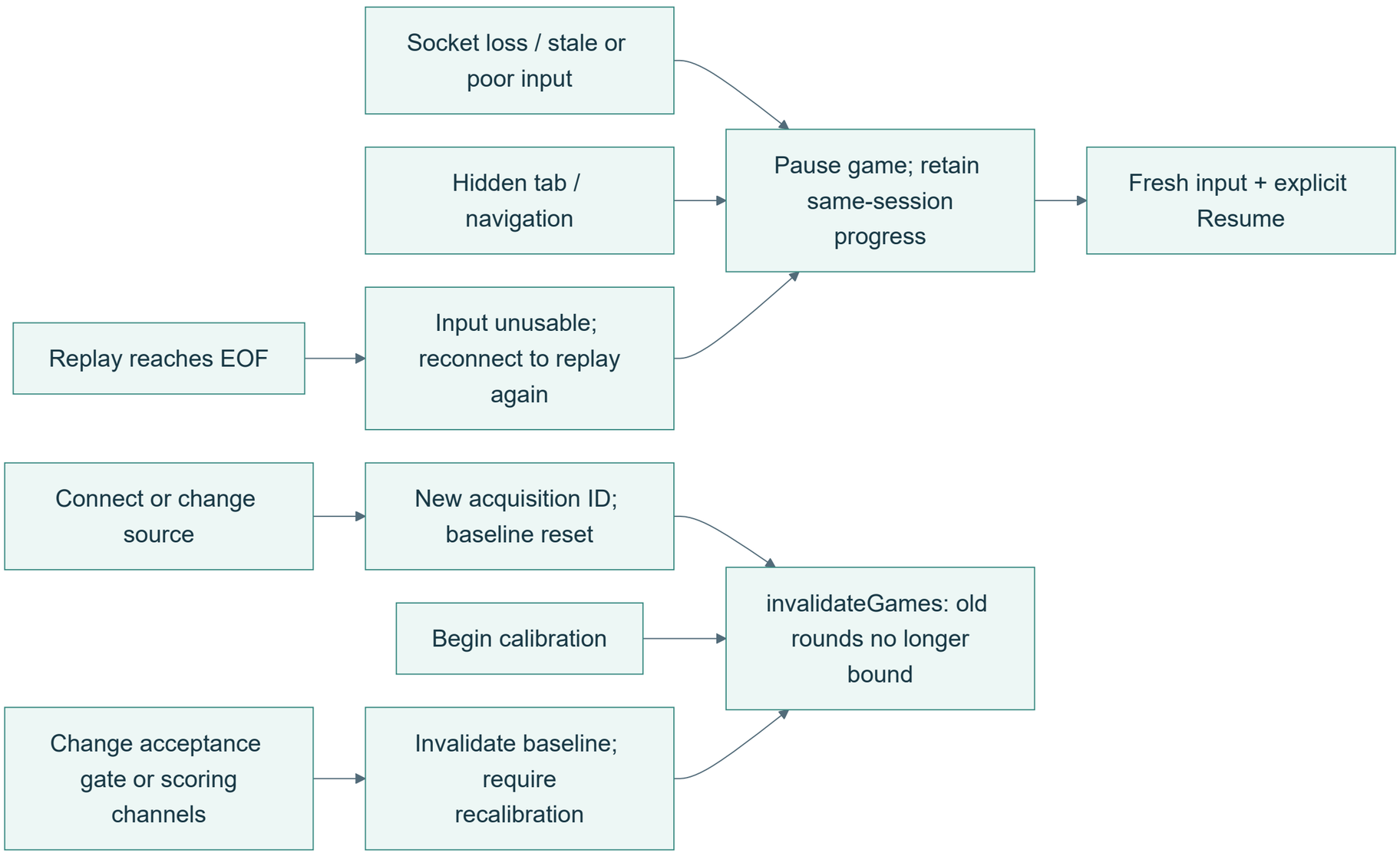
> The map shows a common journey, not a forced page wizard: navigation can open Choose or Visualizer independently. Experiment setup can open without ready input; its Start button has separate raw-EEG requirements. Calibrate is unnecessary for experiments. Opening a game never reconnects or rewinds acquisition.

Per-game lifecycle



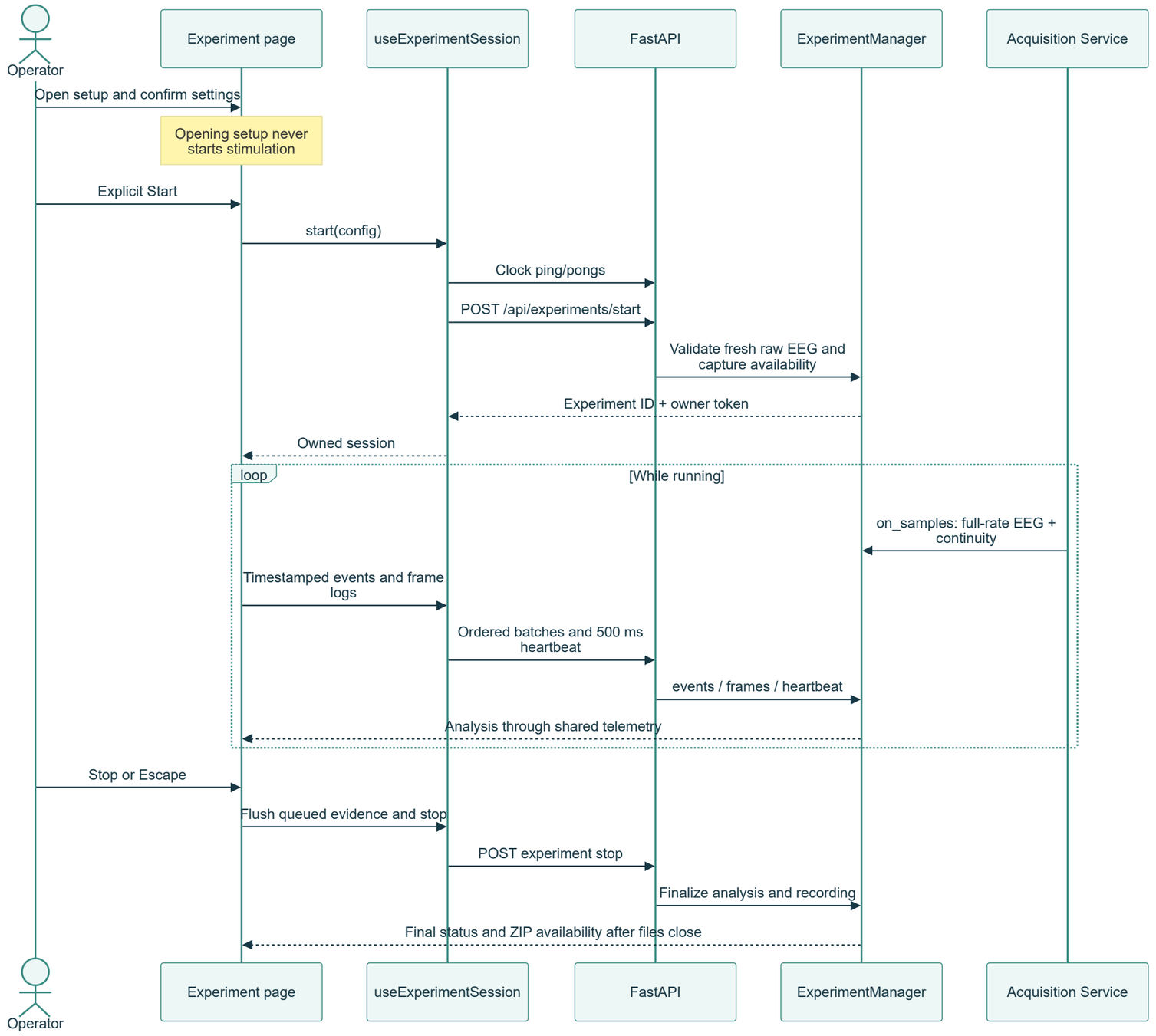
> This is a conceptual lifecycle across the games, not a single enum in the code. A new acquisition or recalibration resets every state to NotStarted; that shared reset path is shown in the interruption map below. Full bloom is a Garden milestone rather than a fall/loss; growth caps at one. A new Garden plant may include the optional settle-in. Cairn and Balance rounds end on falls; a new round resets game state while retaining a valid current session baseline.

What interrupts what



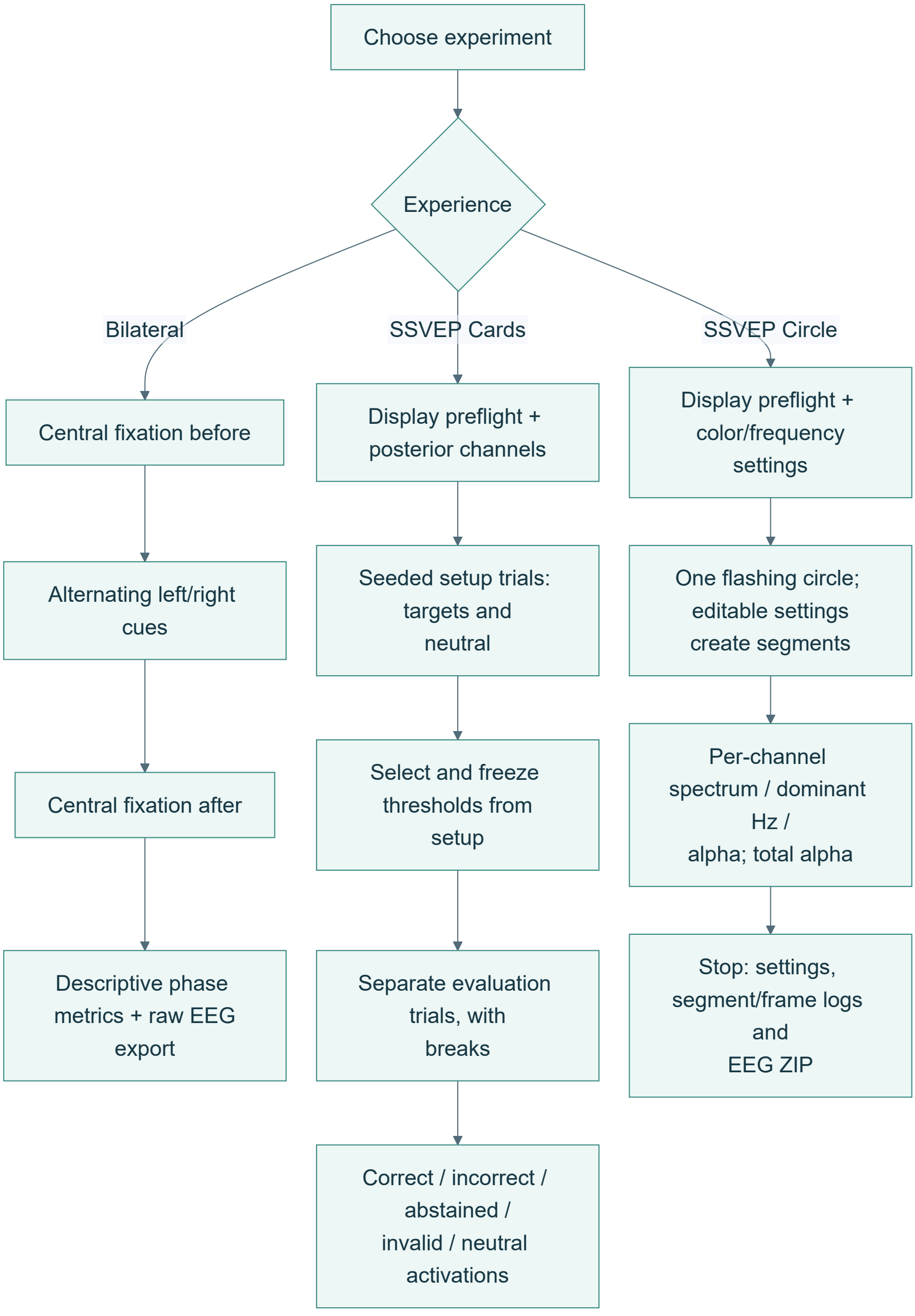
> Research capture can remain active across these game and acquisition transitions. Experiments are more restrictive: source changes, recalibration, leaving the page and other interruption conditions terminate the trial and retain its partial files. See recording and experiment flows.

Shared start and ownership



> Start requires raw input from live, replay or simulated_eeg, not score-only synthetic. Alpha calibration is not required. The UI and backend reject overlap with an active research recording or another experiment. SSVEP pages additionally require a passing display check and acknowledgement; Cards also validates its posterior-channel configuration.

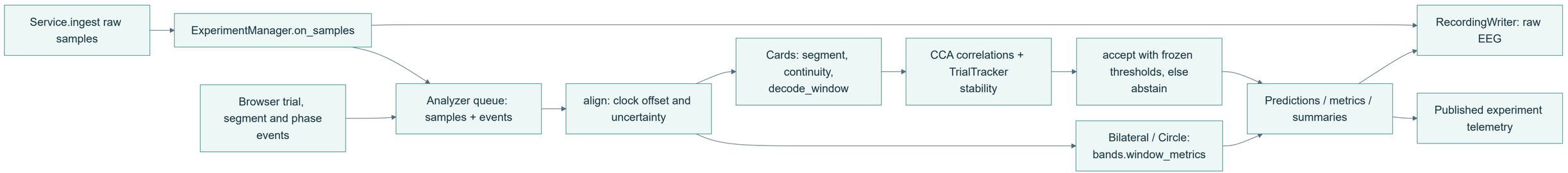
Three different protocols



> Cards supports diagnostic evaluation that abstains when usable thresholds are unavailable; freezing does not guarantee acceptance. Circle is an observation/recording experiment, not a gaze classifier. Bilateral records the chosen follow/fixate instruction condition and descriptive metrics; it is not a P300 classifier or therapeutic assessment.

Source: docs/workspace/maps/flows/experiments.md

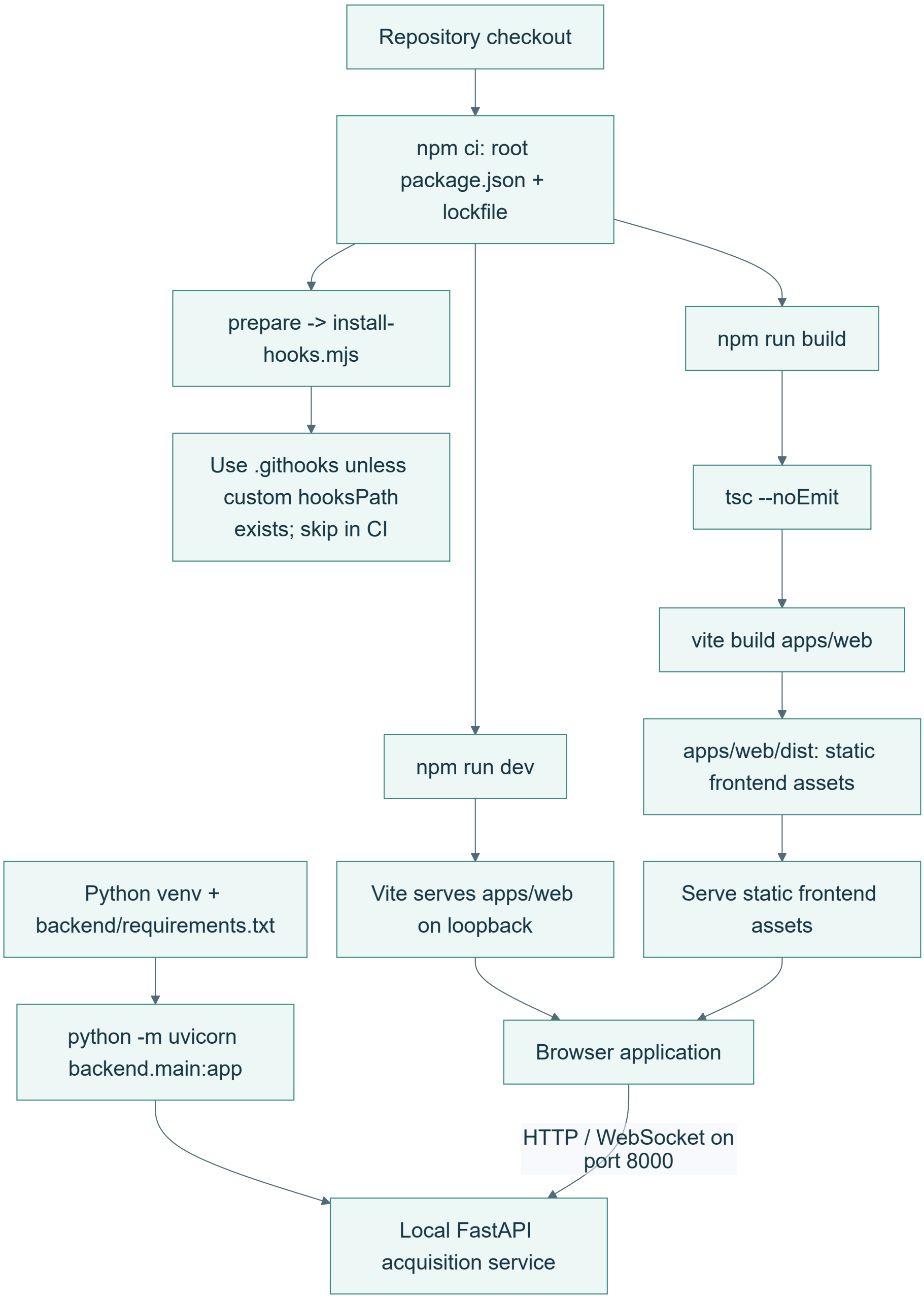
Where decoding happens



In Cards, the instructed target is used for setup/evaluation labels. The decoder computes correlations from raw EEG and frequency references; it does not read the cue as its answer. Replayed resting EEG and generated signals can exercise this software path without validating a new participant response.

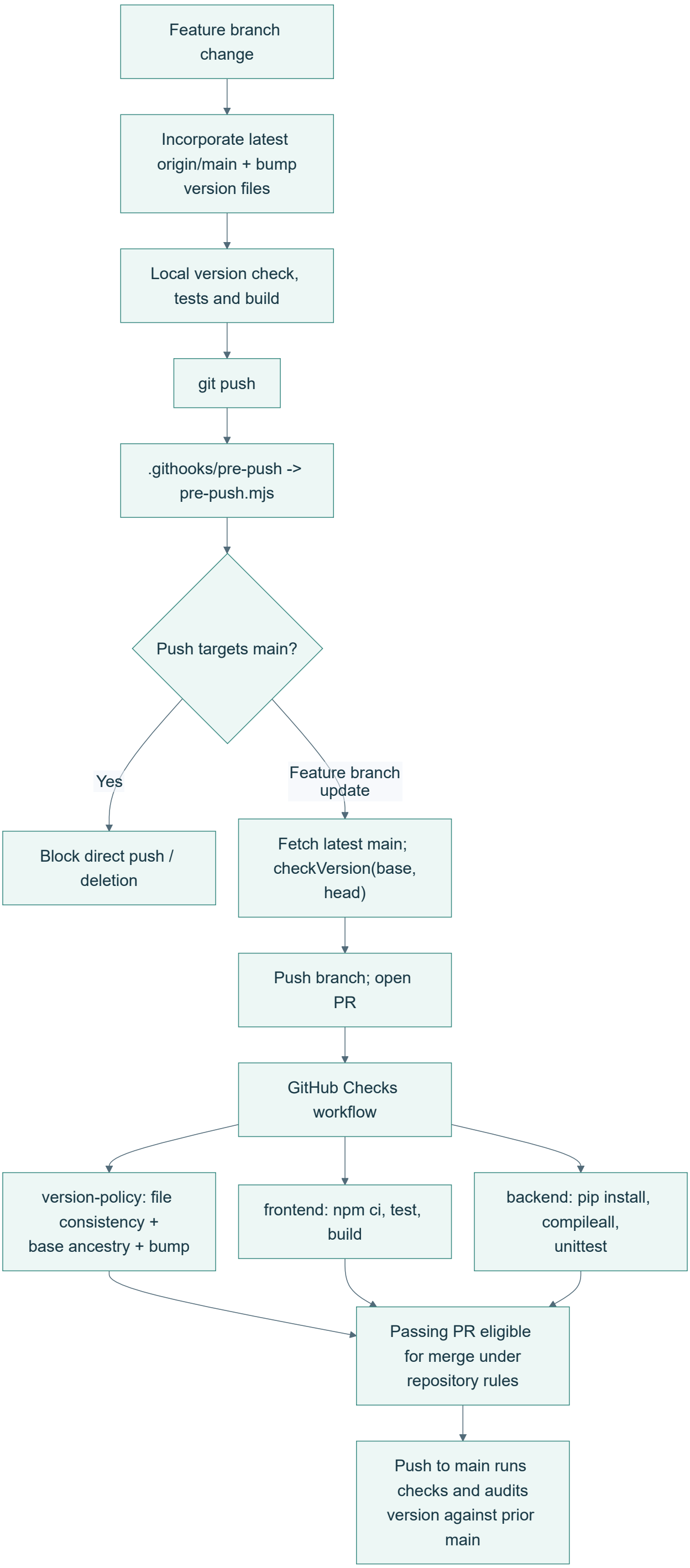
Source: docs/workspace/maps/flows/experiments.md

Install, run and build



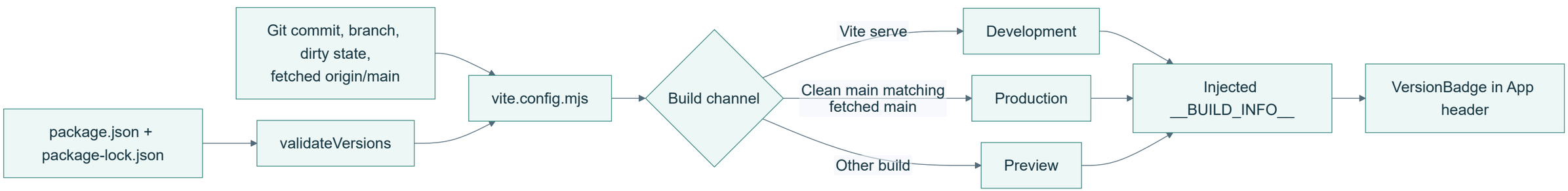
There is one root npm package. The frontend's HTML/source live under apps/web; there is no separate frontend package install. The Python runtime and optional asset-generation tools have separate dependencies. Building the frontend does not launch Python or deploy the app.

Version guard and CI paths



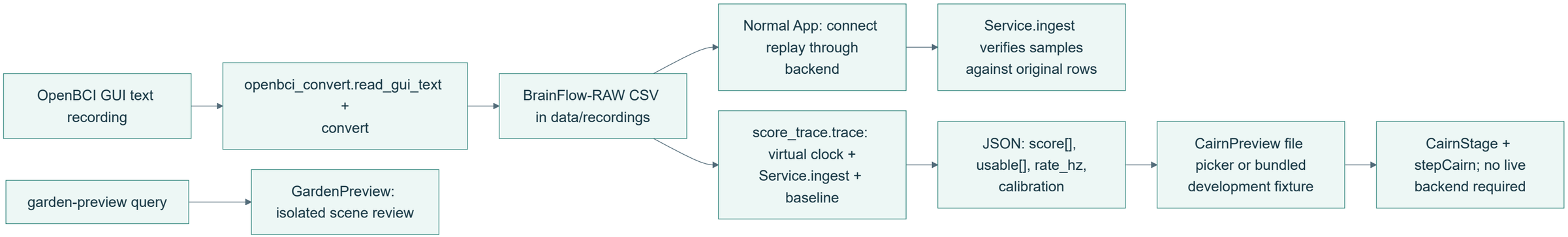
> The hook runs only when installed and is local. The repository includes a GitHub branch-protection configuration for the three checks; the presence of that file does not establish that it has been applied on GitHub. This map describes code/configuration, not a verified remote setting. The workflow contains checks, not a deployment job.

Build information shown in the header



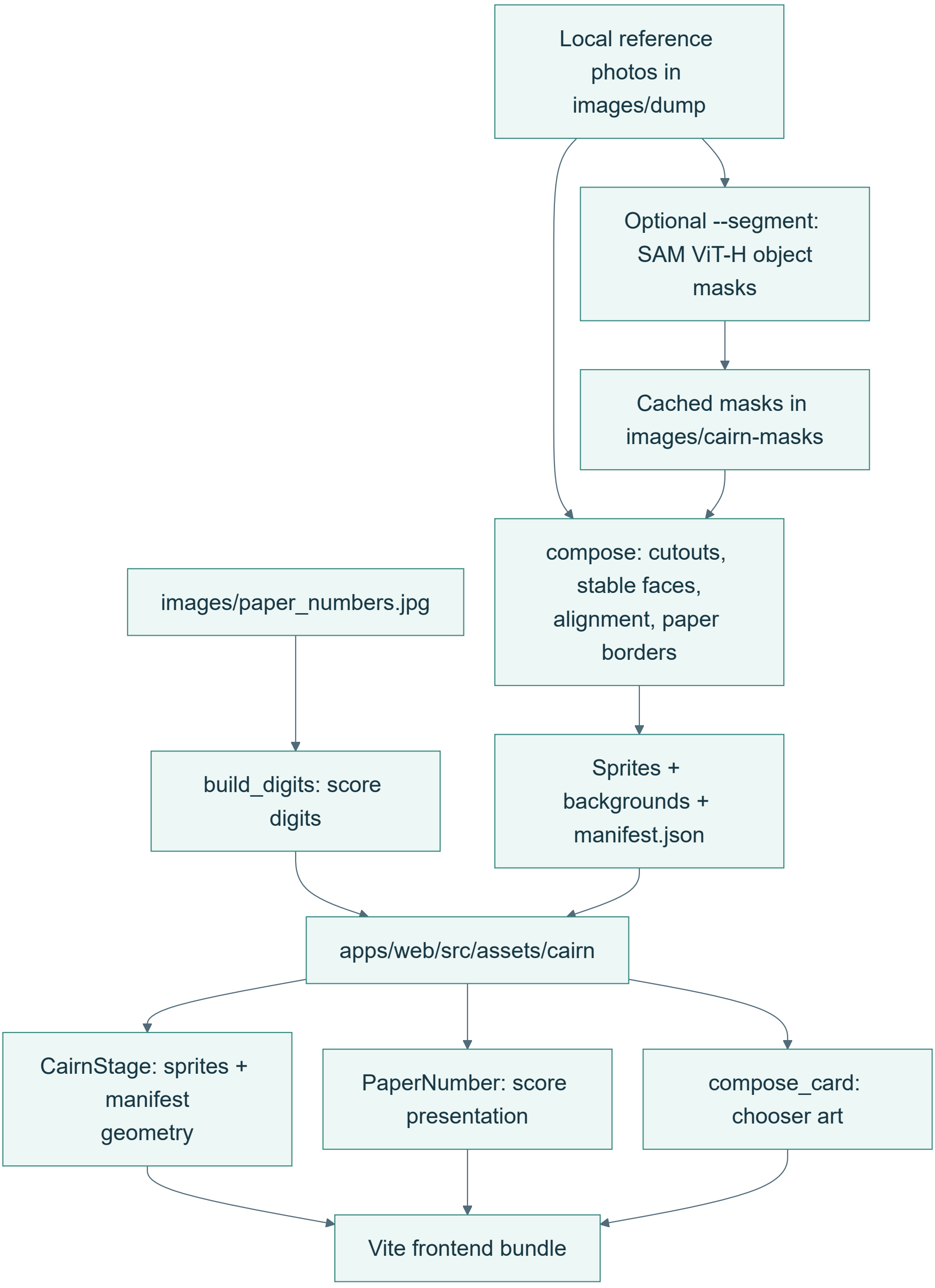
> Metadata is captured when Vite starts/builds. Restart the dev server after changing branch or version. The Production label identifies a source state; it does not publish a build.

Recording conversion and no-headset game testing



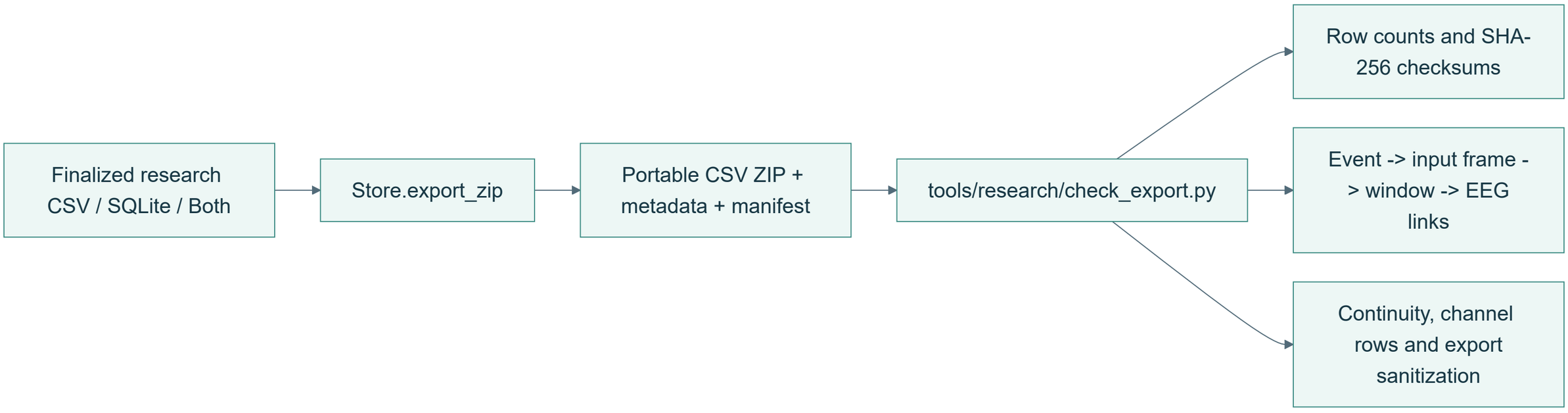
> The converter maps supported GUI text rows to BrainFlow's 24-row Cyton format. score_trace reuses the backend's feature/calibration pipeline with a deterministic virtual clock; it is not a substitute for testing the real board or fresh stimulus responses. Preview routes are enabled only in development.

Cairn photo assets



> The asset script is an on-demand authoring tool, not part of npm run build. Segmentation uses optional heavyweight model tooling; recomposition uses cached masks and image-processing libraries. Its defaults keep original photos under ignored images/dump and write reusable generated assets to the frontend asset directory.

Research export inspection



> Run python tools/research/check_export.py <research-export.zip> against an actual exported recording. The checker inspects recorded structure and consistency; the result does not establish physiological validity or physical stimulus timing.